

Project CC-02: Static NGO Site + CDN Architecture

Infrastructure as Code (Terraform) & Cost Optimization Blueprint

Executive Summary: This document contains the complete infrastructure specification and automated deployment scripts for hosting a high-performance, cost-optimized NGO static web application on Microsoft Azure utilizing Azure Blob Storage and Azure CDN with native SSL/TLS encryption.

1. Infrastructure as Code (Terraform Configuration)

Below is the complete `main.tf` code to provision the entire static website and CDN stack declaratively:

```

provider "azurerm" {
  features {}
}

# 1. Resource Group
resource "azurerm_resource_group" "ngo_rg" {
  name      = "ngo-project-rg"
  location  = "westcentralus"
}

# 2. Storage Account with Static Website Hosting
resource "azurerm_storage_account" "ngo_storage" {
  name                        = "ngostaticstorage2026"
  resource_group_name        = azurerm_resource_group.ngo_rg.name
  location                   = azurerm_resource_group.ngo_rg.location
  account_tier                = "Standard"
  account_replication_type    = "LRS"

  static_website {
    index_document = "index.html"
    error_404_document = "404.html"
  }
}

# 3. Azure CDN Profile
resource "azurerm_cdn_profile" "ngo_cdn_profile" {
  name      = "ngo-cdn-profile"
  location  = "global"
  resource_group_name = azurerm_resource_group.ngo_rg.name
  sku       = "Standard_Microsoft"
}

# 4. Azure CDN Endpoint
resource "azurerm_cdn_endpoint" "ngo_cdn_endpoint" {
  name      = "ngo-cdn-endpoint-2026"
  profile_name = azurerm_cdn_profile.ngo_cdn_profile.name
  location  = "global"
  resource_group_name = azurerm_resource_group.ngo_rg.name

  origin {
    name      = "ngo-static-origin"
    host_name = azurerm_storage_account.ngo_storage.primary_web_host
  }

  origin_host_header = azurerm_storage_account.ngo_storage.primary_web_host
}

```

2. Architectural Cost Breakdown & Financial Note

Component	Service Tier	Estimated Usage	Monthly Cost (USD)
Azure Blob Storage	Standard Hot LRS (\$web)	~5 MB Static Assets	\$0.0001 / mo
Azure CDN	Standard Microsoft	< 5 GB Data Out / mo	\$0.00 - \$0.05 / mo
SSL/TLS Certificate	Managed HTTPS Certificate	1 Custom Domain / Edge	\$0.00 (Free)

Component	Service Tier	Estimated Usage	Monthly Cost (USD)
Total Estimated Cost	Serverless Static Architecture		< \$0.10 / month